

Purposes of Test Norms:-

- ① Relative standing of person (x shows Ranking)
- ② Comparable measure (Permit direct comparison of the individual problems in different test)

Test Norms:

A test norm is a set of scalar data describing the performance of a large number of people on that test. Test norms can be represented by two important statistics: Means and Standard Deviations. The most important measure in psychometrics is the arithmetical average of the mean ($\bar{X}$). Average value

A norm referenced test (NRT) is a type of test assessment or evaluation which yields an estimate of position of tested individual in a predefined population with respect to traits being measured. This estimate is derived from the analysis of test score and possibly other relevant data from a sample drawn from population. (Sum of data or score)

Types of test norms:

Fundamentally, derived scores are expressed in these major ways

- Development Norms
- Within group Norms.
- National Norms
- Norm-Group

1. Developmental Norms: e.g. (CIT) - (Cross intelligent Test) - To evaluate Mental Age equivalent in MR people. These type of norms generally indicate the normal developmental path the individual has progressed. They are very helpful for descriptive purpose but they are not compatible to precise statistical treatment. The types of development norms are

- (a) Mental Age Norms (MAN)
- (b) Grade Equivalent Norms (GEN)
- (c) Ordinal Scale Norms (OSN)

2. Within Group Norms: e.g. SPM - for normal individual.

Such type of norms help in comparing the individual's performance with the most nearly comparable standardized group's performance. Within group norms have a uniform and clearly defined quantitative meaning and can be appropriately employed in most types of statistical analyses.

- Percentiles: ↓ = Poor rank ↓
Percentile scores represent the percentage of persons in the standardized sample who fall below a given raw score. They indicate an individual's relative position in the standardized sample. In case of percentiles, the counting begins from the bottom so lower the percentile, poorer the standing rank.

Developmental norms are defined as standards by which the progress of a child's development can be measured.

For Example

The average age at which child walks, learn to talk or reaches puberty would be such a standard and would be used to judge whether the child is progressing normally.

⇒ Norms always preset / pre determined
⇒ pre criteria.

Standard Score:

Standard score express the individual's distance from the Mean in terms of the standard deviation of the distribution. They are obtained by linear or nonlinear transformation of the original raw scores. T Scores and Z scores are known as standard scores.

Age Norms:

To establish age norms, the Mean of raw scores obtained by all in the same age group within a standardized sample is taken. So mean raw score of 12 year old students would represent the 12 years norm.

Grade Norms:

Grade norms are found by computing the Mean raw score obtained by students in particular grade.

3. National Norms:

Most nationally distributed tests are provided with some sort of "national" norms. A nationwide norms population has certain advantages:

- The population can be readily and clearly defined for the test score consumer
- Most examinees tested are likely to be members of this group, or of some presumably very similar group following a year or two later
- Such a group is roughly equally appropriate for publishers in various parts of the country, thus in theory making test norms comparable from publisher to publisher.

4. Norm Group: is a collection of scores, from a relevant population

It is desirable that a set of test norms should be based on a group that can be easily definitively described, is familiar to the consumer of test scores and includes as one of its members the examinee whose test score is to be interpreted which allows you to benchmark i.e. if you're looking at graduates.

Cut-off scores:

A cutoff score is an established score used to filter out unqualified candidates on any particular test or assessment. There are several criteria for determination of the most appropriate cut-off value in a diagnostic test with continuous results

example: In employment testing, cut-off scores can be established to filter out job candidates who did not score high enough on a particular pre-employment test.

⇒ Many organizations opt to not use a "hard" cut-off score, meaning they will still consider applicants who do not score in the desired range. If a hard cut-off score is used, deciding where to set a cut-off score depends on a number of different factors. In order to establish a reasonable cut-off score for a particular position, a company can administer the test to its current employees in that position, and then establish a cut-off score based on the current employees' scores. This technique for establishing cut-off scores requires a large pool of test takers, so companies that do not already have a large number of current employees in that position can instead rely on standard cut-off score recommendations provided by testing companies.

National Norms are standardized test designs to compare & rank test takers in relation to one another using national standards.

categories of Testing

Norm Referenced Testing
Criterion Referenced Tests

② Another factor to consider when establishing cut-off scores is the size of the applicant pool. If a company is faced with a rather large applicant pool for a particular position, it can afford to set a reasonably high cut-off score. Setting higher cut-off scores not only makes it more likely that the candidates will be qualified, but it also helps reduce the amount of labour involved in the hiring process by narrowing the list of qualified candidates to move through the hiring process. However, if a company is dealing with smaller applicant pools or needs to hire a lot of people for a particular position, setting high cut-off scores may unnecessarily constrict the list of qualified applicants, making it more difficult to find and hire employees for that position.

Norm referenced testing (NRT)

1. Norm-referenced tests are a form of standardized testing that compares "normal" skill levels to those of individual students of the same age. By comparing students to one another, it is possible to determine whether, how, and to what a degree a particular student is ahead of or behind the norm. These tests help to diagnose learning disorders and also help special education teachers and other professionals develop appropriate program planning for students with disabilities.

Norm-referenced tests are developed by creating the test items and then administering the test to a group of students that will be used as the basis of comparison. Statistical methods are used to determine how raw scores will be interpreted and what performance levels are assigned to each score.

A norm-referenced test (NRT) refers to a test on which the score of an examinee is reported in comparison to a distribution of scores of other examinees in a reference group. The standards for comparisons are called norms, and the group in which the norms are obtained is called a norm group. Norm-referenced testing differs from criterion-referenced testing (CRT), in that the primary purpose of CRTs is to determine how examinees perform in comparison to a predetermined performance level or outcome. NRTs provide information on how well an examinee performs in comparison to other examinees, whereas CRTs provide information on what an examinee knows and can do. A mathematical test used to determine an examinee's performance against specific curriculum outcomes is an example of CRT. By comparison, when assessing mathematical aptitude of examinees, the reference point becomes a specific comparison group.

② Criterion-referenced tests: (For special skills Determination)

1. These are designed to find out whether a child has a set of skills, rather than how a child compares to other children of the same age (normed tests). The test designers analyze the component parts of specific academic skills, such as number understanding, and then write test items that will measure whether the child has all the component parts of the skill. The tests are normed in terms of what skill level a child should have. Still, the tests are designed to measure a child's acquisition of specific skills.

A test of reading skills would seek to discover whether a child can identify the specific sounds consonants make before it would evaluate whether a student can answer comprehension questions. The questions in a criterion-referenced test seek to find if the

→ Score
→ S.C %
→ Ranking of ind.
→ Comparing student
→ Ex: SAT, GRE

→ % of item to ans correctly
→ Student master skills/knowledge
→ Achievement
→ Exams, Chapter Test

student has the skills, not whether the student does as well as other third grade children. In other words, a criterion-referenced test will provide important information that a teacher can use to design specific instructional strategies to help those students succeed.

It will identify skills that the students lack.

A criterion-referenced test for Mathematics should reflect the scope and sequence of state standards (such as the common core state standards.) It would reflect the skills needed at each age, for young mathematicians, understanding one to one correspondence, numeracy and at least addition as an operation. As a child grows, they are expected to gain new skills in a reasonable order that builds on earlier levels of skill acquisition.

State high stakes tests of achievement are criterion-referenced tests that are aligned with the state's standards, measuring whether children actually have mastered the skills that are prescribed for the students' particular grade-level. Whether these tests are actually reliable or valid may or not be true: unless the test designer has actually compared the success of students (say in reading new texts, or succeeding in college) with their "scores" for the test, they may not actually be measuring what they claim to measure.

Examples

The Key Math Tests are criterion-referenced achievement tests that provide both diagnostic information and achievement scores in math.

Other criterion-referenced tests include the Peabody Individual Achievement Test (PIAT) and the Woodcock Johnson Test of Individual Achievement.

Computer-assisted testing (CAT):

Computer-assisted testing (CAT) in counseling and therapy is becoming increasingly common due to dramatic improvements in cost-effectiveness and increased counselor familiarity with computer applications. The assumption underlying the use of CAT is that the effectiveness of counseling is improved by allocating repetitive computational and instructional tasks to the computer, thus allowing counselors to more fully focus on interpersonal tasks, such as helping clients understand the role of testing in counseling and helping clients integrate the self-knowledge obtained in testing into a concrete plan for behavior change. The potential benefits of CAT, however, need to be evaluated against the potential limitations of this technology.

Computer-Assisted Testing Options

The following options exist for using computer-assisted testing in counseling and therapy:

1) Test administration via: a) keyboard input by the client from items presented on the computer display, with alternative input options available for physically disabled clients; or b) client

completion of a specially prepared test answer sheet that is then optically scanned for computer input; or c) client completion of a traditional test answer sheet with keyboard input by a clerical staff person.

2) Test scoring via the computer (local or remote).

3) Test score profile generation via the computer.

4) Narrative interpretive report generation via the computer with reports available for both the client and practitioner if appropriate (the narrative report may also include the test profile).

5) Videodisc-based generalized test interpretation provided to the client immediately following test administration (Sampson, 1990a, p 452-453).

AGE NORMS: refers to typical developmental milestones, behaviors or abilities that are expected of individuals at certain ages within a population. These norms are based on empirical research and observations of typical patterns of development. They serve as benchmarks for assessing whether individuals are progressing normally in various areas of development.

Examples:

Examples of age norms include:

1- LANGUAGE DEVELOPMENT:

By age 2, most children can use simple words and phrases to communicate basic needs & desires. By age 5, they can typically speak in full sentences & engage in conversations with others.

2- MOTOR SKILLS:

By age 3, children can usually walk, run and climb stairs with assistance. By age 6, they can typically ride a bicycle, tie their shoes and catch a ball with both hands.

below a given raw score. They indicate an individual's relative position in the standardized sample. In case of percentiles, the counting begins from the bottom so lower the percentile, poorer the standing rank.

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